

Implementation of Puppet Classes for Reusable Configuration Management

Problem Scenario

An organization is managing multiple Linux servers using Puppet.

The administrator needs to configure an application directory called `myapp` on Puppet-managed machines. The configuration requires:

- Creating `/home/Man`
- Creating `/home/Man/myapp`
- Creating a `config.txt` file
- Maintaining specific content inside the configuration file
- Ensuring that the directories are created before the file

Initially, all these resources were written directly inside `site.pp`.

As the Puppet infrastructure grows, putting all configuration logic into `site.pp` becomes difficult to maintain.

Requirement

Create a **Puppet class** named `myapp` that groups all application-related resources into a reusable configuration unit, and invoke this class from `site.pp`.

Prerequisites

Before performing this practical, ensure:

Puppet Master

```
sudo systemctl status puppetserver
```

should show:

Active: active (running)

Puppet Agent

Puppet Agent should be able to communicate with the Master:

```
sudo /opt/puppetlabs/puppet/bin/puppet agent --test
```

Your Master-Agent communication is already working.

On Master:

```
ubuntu@puppet-master:~$ sudo nano /etc/puppetlabs/code/environments/production/manifests/site.pp
ubuntu@puppet-master:~$ sudo cat /etc/puppetlabs/code/environments/production/manifests/site.pp
    include myapp
ubuntu@puppet-master:~$
```

```
ubuntu@puppet-master:~$ sudo nano /etc/puppetlabs/code/environments/production/manifests/myapp.pp
ubuntu@puppet-master:~$ sudo cat /etc/puppetlabs/code/environments/production/manifests/myapp.pp
class myapp {

    file { ['/home/Man':
        ensure => directory,
    }

    file { ['/home/Man/myapp':
        ensure => directory,
        require => File['/home/Man'],
    }

    file { ['/home/Man/myapp/config.txt':
        ensure => file,
        content => "Application managed by Puppet\n",
        require => File['/home/Man/myapp'],
    }

}
ubuntu@puppet-master:~$
```

On Agent:

```
ubuntu@puppet-agent:~$ sudo /opt/puppetlabs/puppet/bin/puppet agent --test
Info: Refreshing CA certificate
Info: CA certificate is unmodified, using existing CA certificate
Info: Refreshing CRL
Info: CRL is unmodified, using existing CRL
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Warning: Fact value '-----BEGIN CERTIFICATE-----
```

```
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
Notice: Requesting catalog from puppet-master:8140 (172.31.3.190)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent@enath-1.compute.internal
Info: Applying configuration version '1789835573'
Notice: /Stage[main]/Myapp/File[/home/Man/myapp/config.txt]/content:
--- /home/Man/myapp/config.txt 2026-09-19 16:23:44.631572469 +0800
+++ /tmp/puppet-file20260919-1408-1tqta 2026-09-19 16:39:33.437297720 +0800
@@ -1,1 @@
+Application managed by puppet
+Application managed by Puppet
Notice: /Stage[main]/Myapp/File[/home/Man/myapp/config.txt]/content: content changed '[sha256]65d3b748b1c17f2e2f97002b6bb5d2ad8189b8a296e3dec30c03b0446f9dd74f' to '[sha256]3597956821f37d15d2218cf922d3b12e0a7be244d4ade21b0b17dafa4787cd'
Notice: Applied catalog in 0.03 seconds
ubuntu@puppet-agent:~$
```